Xinxu Wei

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Target Roles: Research Scientist | Applied Scientist | ML Engineer
Ph.D. Student, Lehigh University

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Availability: Seeking full-time internship opportunities from Sep 2026 to Dec 2026. Graduating in Dec 2026 and available for full-time roles starting in Dec 2026 or Jan 2027.

Education

•Lehigh University *2023.08 - 2026.12 (Expected)*
Ph.D. in Computer Science. Advisor: Prof. Yu Zhang and Prof. Lifang He Bethlehem PA, USA

•McGill University (Dual Master Degree) *2021.08 - 2023.05*
M.Sc. in Biomedical Engineering. Advisor: Prof. Danilo Bzdok Montreal QC, Canada

•University of Electronic Science and Technology of China (UESTC) *2020.08 - 2023.05*
M.Eng. in Biomedical Engineering. Advisor: Prof. Yongjie Li Chengdu, China

•University of Electronic Science and Technology of China (UESTC) *2016.08 - 2020.05*
B.Eng. in Computer Engineering Chengdu, China

Professional Summary

- Ph.D. candidate in Computer Science (expected Dec 2026) focused on graph foundation models, multimodal learning, and medical AI for neuroscience.
- First-author publications at top-tier venues, including ICLR 2026 and ICML 2025; additional papers in Neural Networks, ESWA.
- Designed novel pre-training paradigms, built foundation models for brain graph data, and open-sourced end-to-end training and evaluation frameworks.
- Active reviewer for top-tier conferences and journals; Google Scholar citations: 200+.

Research Areas

Graph Foundation Models: Graph Neural Networks (GNNs), Hyper-Graphs, Graph Pre-training, Graph Prompt/Few-shot Transfer, Graph Contrastive and Masked Modeling
Representation Learning: Self-supervised Learning, Contrastive Learning, Pre-training/Fine-tuning, Meta Learning, Transfer Learning, Knowledge Distillation
LLMs, Agents, and Recommendation: LLM Fine-tuning, LLM Agents, Multi-agent Systems, Tool Use/Function Calling, Retrieval-Augmented Generation (RAG), Agentic Workflows, User-Item Representation Learning, Retrieval and Ranking
Medical AI and Vision: Brain Graph Modeling, fMRI/EEG Fusion, Image Segmentation, Image Enhancement, Vision-Language Modeling

Technical Skills

Programming Languages: Python, C/C++, Java, Bash, Matlab
ML Frameworks: PyTorch, TensorFlow, Keras, DGL, PyG
Libraries: Numpy, Scikit-Learn, Pandas, Matplotlib, Seaborn, OpenCV, Scipy, Transformers, HuggingFace
ML Engineering Tools: Git, GitHub, Docker, Linux Shell, Jupyter Notebook, Conda, VSCode, Vim
Modeling: CNN, RNN/LSTM/GRU, GCN/GAT, Autoencoders, Variational Autoencoders (VAE), Capsule Networks
Soft Skills: Technical Writing, Research Communication, Cross-team Collaboration, Project Management

Internship

•TikTok *2026.05 - 2026.08*
Incoming Research Scientist Intern Seattle, WA, USA

- Incoming intern on recommendation systems to improve retrieval/ranking quality with large-scale modeling and offline-online evaluation workflows.
- Planned work includes graph-based user-item representation learning to capture higher-order behavioral signals for personalization.
- Will explore LLM-enhanced semantic understanding and model improvements for robust recommendation performance.

Open-Source Projects

- **BrainGFM [Code] [Paper]**: Built a brain graph foundation model with large-scale pre-training and prompt-tuning across atlases and disorders. Implemented end-to-end data processing, model training, and transfer evaluation pipelines for reproducible few-shot/zero-shot adaptation.
- **EEG-DisGCMAE [Code] [Paper]**: Built an EEG foundation model pre-training framework with graph contrastive masked autoencoding and teacher-student distillation. Developed pre-training and fine-tuning workflows to transfer representations from high-density EEG to low-density EEG settings.
- **OCE-Net [Code] [Paper]**: Developed an orientation-context entangled segmentation model for retinal vessel analysis. Provided training/inference code and benchmarking scripts for reproducible evaluation on medical image segmentation tasks.

Selected Publications

- 🏆 **X. Wei**, K. Zhao, Y. Jiao, L. He, and Y. Zhang. **A Brain Graph Foundation Model: Pre-Training and Prompt-Tuning across Broad Atlases and Disorders.** International Conference on Learning Representations (ICLR), 2026, Accepted. (CCF A, Core A*), **[OpenReview] [Paper] [Code]**
- 🏆 **X. Wei**, K. Zhao, Y. Jiao, H. Xie, L. He, and Y. Zhang. **Pre-Training Graph Contrastive Masked Autoencoders Are Strong Distillers For EEG.** International Conference on Machine Learning (ICML), 2025, Accepted. (CCF A, Core A*) **[OpenReview] [Paper] [Code]**
- 🏆 **X. Wei**, R. Zhou, L. He, and Y. Zhang. **Diffusion-Guided Pretraining for Brain Graph Foundation Models.** Under Review by International Conference on Machine Learning (ICML), 2026. (CCF A, Core A*) **[Paper]**
- **X. Wei**, K. Zhao, Y. Jiao, N. B. Carlisle, H. Xie, G. A. Fonzo, and Y. Zhang. **Multi-Modal Cross-Domain Self-Supervised Pre-Training For fMRI And EEG Fusion.** Neural Networks, 2025. (JCR Q1, CCF B, IF=6.4) **[Paper]**
- **X. Wei**, K. Yang, D. Bzdok, and Y. Li. **Orientation And Context Entangled Network For Retinal Vessel Segmentation.** Expert Systems with Applications, 2023. (JCR Q1, IF=7.5) **[Paper] [Code]**
- **X. Wei**, X. Zhang, and Y. Li. **DA-DRN: Degradation-Aware Deep Retinex Network For Low-Light Image Enhancement.** Digital Signal Processing, 2024. (JCR Q2, IF=3.0) **[Paper]**
- **X. Wei**, X. Niu, X. Zhang, and Y. Li. **Deep Pneumonia: Attention-Based Contrastive Learning For Class-Imbalanced Pneumonia Lesion Recognition In Chest X-Rays.** In Proceedings of the IEEE International Conference on Big Data (BigData), 2022. (Core B) **[Paper]**
- **X. Wei**, X. Zhang, and Y. Li. **TSN-CA: A Two-Stage Network With Channel Attention For Low-Light Image Enhancement.** In Proceedings of the International Conference on Artificial Neural Networks (ICANN), 2022. (Core B) **[Paper]**
- **X. Wei**, X. Zhang, and Y. Li. **SARN: A Lightweight Stacked Attention Residual Network For Low-Light Image Enhancement.** In Proceedings of the International Conference on Robotics and Automation Engineering (ICRAE), 2021. **[Paper] [Code]**
- **X. Wei**, X. Lin, and Y. Li. **Retinal Vessel Segmentation With Deep Graph And Capsule Reasoning.** Submitted to IEEE Transactions on Artificial Intelligence (TAI), under review, 2024. **[Paper]**
- X. Lin, **X. Wei**, S. Zhao, and Y. Li. **Vascular Skeleton Deformation Evaluation Based On The Metric Of Sinkhorn Distance.** In Proceedings of the IEEE International Symposium on Biomedical Imaging (ISBI), 2024. (Core B) **[Paper] [Code]**
- Y. Jiao, K. Zhao, **X. Wei**, N. Carlisle, C. Keller, D. Oathes, G. Fonzo, and Y. Zhang. **Deep Graph Learning Of Multimodal Brain Networks Defines Treatment-Predictive Signatures In Major Depression.** Molecular Psychiatry, 2025. (JCR Q1, IF=9.6) **[Paper] [Code]**
- Y. Jiao, **X. Wei**, L. He, and Y. Zhang. **A Functional System-Informed Graph Neural Network Framework To Quantify Interpretable Brain Dysfunction In ASD.** Neural Networks, 2025. (JCR Q1, CCF B, IF=6.4) **[Paper]**
- X. Lin, **X. Wei**, A. Shmuel, and Y. Li. **VPBSD: Vessel-Pattern-Based Semi-Supervised Distillation For Efficient 3D Microscopic Cerebrovascular Segmentation.** Submitted to Pattern Recognition, 2024. **[Paper]**

Experience & Projects

•Lehigh University *2023.08 - Present*

Research Assistant (Completed 5 research papers, including 3 as the first author and 2 as a co-author) Bethlehem PA, USA

- Built large-scale fMRI brain graph/hyper-graph datasets and integrated diffusion-based kernels to model long-range and higher-order dependencies for pre-training. **[Under Review by ICML 2026]**.
- Developed a brain graph foundation model pipeline (pre-training, prompt tuning, and transfer) for cross-disorder and cross-atlas adaptation. **[ICLR 2026 Accepted]**.
- Proposed an EEG graph pre-training framework with knowledge distillation from high-density to low-density EEG settings. **[ICML 2025 Accepted]**.
- Built multimodal fMRI+EEG self-supervised pipelines for fusion and transfer, contributing to publications in **[Molecular Psychiatry 2025]** and **[Neural Networks 2025]**.

•Mila - Quebec AI Institute *2022.01-2022.08*

Research Assistant (Completed 5 research papers, including 3 as the first author and 2 as a co-author) Montreal QC, Canada

- Designed orientation-aware dynamic convolution and imbalance-aware attention for retinal vessel segmentation. **[ESWA 2023 Accepted]**.

– Extended segmentation modeling with graph and capsule reasoning, and developed vessel-structure evaluation and representation methods for follow-up works. **[IEEE TAI Submission] [ISBI] [PR Submission]**.
– Built a self-supervised contrastive pre-training pipeline for pneumonia recognition on chest X-rays. **[IEEE BigData 2022 Accepted]**.

•UESTC *2020.08-2023.05*

Research Assistant (Completed 3 first-author research papers and 2 patents) Chengdu, China

– Developed low-light image enhancement models, including plug-and-play and channel-aware designs, with resulting publications at **[IEEE ICRAE 2021]**, **[ICANN 2022]**, and **[DSP 2024]**.
– Built lightweight face analysis systems for fatigue detection and facial emotion recognition, resulting in two granted patents **[Patent 1] [Patent 2]**.

Academic Services

- **Conference Reviewer**: NeurIPS 2025, ICML 2025/2026, ICLR 2024/2025/2026, KDD 2025, AAAI 2024/2025, IJCAI 2025, MICCAI 2025, ISBI 2023, ICPR 2022
- **Journal Reviewer**: IEEE Transactions on Neural Networks and Learning Systems (TNNLS), IEEE Transactions on Knowledge and Data Engineering (TKDE), IEEE Transactions on Circuits and Systems for Video Technology (TCSVT), IEEE Transactions on Medical Imaging (TMI), IEEE Transactions on Artificial Intelligence (TAI), Pattern Recognition (PR), Computer Vision and Image Understanding (CVIU), Neural Networks (NN), Expert Systems with Applications (ESWA), Knowledge-Based Systems (KBS), Engineering Applications of Artificial Intelligence (EAAI), Signal, Image and Video Processing (SIVP), Digital Signal Processing (DSP), Signal Processing

Honors and Awards

- 2023.08 Lehigh University Fellowship
- 2023.05 National Scholarship of China
- 2022.05 First-Class Scholarship of UESTC
- 2022.04 Suzhou Industrial Scholarship